

A Free-Space Pair Counterexample for BAP of Banach Pairs

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2026-06-20

Status

This packet gives a likely valid counterexample to one natural reading of the Section 7 question in Chavez-Dominguez, *An Ando-Choi-Effros lifting theorem respecting subspaces*, arXiv:1702.04774. For Banach-space pairs, BAP of the Lipschitz-free pair $(\mathcal{F}(X), \mathcal{F}(Y))$ does not imply BAP of the original pair (X, Y) .

Source Passage

The source first gives a separable pair with Lipschitz BAP but not BAP, and then asks whether an analogue of the Godefroy-Kalton free-space result holds for pairs:

$\|TP(x) - P(x)\| \leq \varepsilon/2$ and $\|S(x - P(x)) - (x - P(x))\| \leq \varepsilon/2$. Note that in addition we may assume $S(0) = 0$. Now consider the map $R = TP + S(Id_X - P) : X \rightarrow X$. Clearly R has finite-dimensional range, it has Lipschitz constant at most $\text{Lip}(T) \text{Lip}(P) + \text{Lip}(S) \text{Lip}(Id_X - P) \leq \mu \text{Lip}(P) + \lambda(1 + \text{Lip}(P))$, for any $y \in Y$ we have $R(y) = T(y) + S(0) = T(y) \in Y$, and for every $x \in K$

$$\begin{aligned} \|R(x) - x\| &= \|TP(x) + S(x - P(x)) - x\| \\ &\leq \|TP(x) - P(x)\| + \|S(x - P(x)) - (x - P(x))\| < \varepsilon. \end{aligned}$$

□

We are now ready for the example.

Proposition 6.11. *There exists a separable pair of a Banach space and a subspace (X, Y) with the Lipschitz BAP, but without the BAP. In particular, (X, Y) does not have the Lipschitz-lifting property.*

Proof. Let Z be a separable Banach lattice without the Approximation Property [Sza76], and let (E_n) be a paving of Z . Let $X = c(E_n)$ and $Y = c_0(E_n)$; clearly X and Y are separable and have the BAP. In particular, they both have the Lipschitz BAP. By [Kal12, Thm. 5.2], there is a Lipschitz retraction from $Y^{**} = \ell_\infty(E_n)$ onto Y . Restricting this map to $X = c(E_n)$ gives a Lipschitz retraction from X onto Y , and now it follows from Proposition 6.10 that the pair (X, Y) has the Lipschitz BAP. If the pair (X, Y) had the BAP then X/Y would have the BAP as well by [FJP11, Cor. 1.2], but X/Y is isometric to Z . The last part of the conclusion now follows from Theorem 6.9. □

7. BAP FOR PAIRS OF LIPSCHITZ-FREE SPACES

The Godefroy-Kalton theorem [GK03, Thm. 5.3] not only shows the equivalence between the BAP and the Lipschitz BAP for a Banach space X , but also that these properties are equivalent to the BAP for the corresponding Lipschitz-free space $\mathcal{F}(X)$. We do not know whether a similar result holds for the BAP for pairs. Nevertheless, our Ando-Choi-Effros theorem respecting subspaces can be used to characterize the BAP for pairs of free spaces over compact metric spaces. The result is a version for pairs of [God15, Thm. 2.1]. Before stating it, let us introduce some notation. If $K \subset M$ are metric spaces, we will always assume that they share the same distinguished point whenever we consider their associated

Proof Intuition

The source's Proposition 6.11 already supplies a pair (X, Y) that has too much nonlinear approximation and too little linear approximation: it has the Lipschitz BAP but not the BAP. The extra useful feature in the proof is a Lipschitz retraction $P : X \rightarrow Y$. Passing to free spaces linearizes this retraction into a bounded projection $\widehat{P} : \mathcal{F}(X) \rightarrow \mathcal{F}(Y)$. Meanwhile, the source also notes that X itself has the BAP, so the Godefroy-Kalton theorem gives the BAP for $\mathcal{F}(X)$. A complemented subspace of a BAP space gives a BAP pair by a direct block-diagonal approximation argument. Therefore the free-space pair has BAP even though the original pair does not.

Definitions

For a Banach space X and a closed subspace Y , the pair (X, Y) has the BAP if there is a uniformly bounded net of finite-rank operators $T_i : X \rightarrow X$ converging to the identity uniformly on compact subsets of X and satisfying $T_i(Y) \subseteq Y$ for every i .

If M is a pointed metric space, $\mathcal{F}(M)$ denotes its Lipschitz-free space. A Lipschitz map $f : M \rightarrow N$ fixing the base point has a canonical bounded linearization $\widehat{f} : \mathcal{F}(M) \rightarrow \mathcal{F}(N)$ with $\widehat{f}(\delta_m) = \delta_{f(m)}$ and $\|\widehat{f}\| = \text{Lip}(f)$.

A Complemented-Pair Lemma

Lemma 1. *Let E be a Banach space with the BAP, and let $W \subset E$ be the range of a bounded projection $Q : E \rightarrow E$. Then the pair (E, W) has the BAP.*

Proof. Choose uniformly bounded finite-rank operators $A_i : E \rightarrow E$ converging to Id_E uniformly on compact subsets. Put

$$R_i = QA_iQ + (\text{Id} - Q)A_i(\text{Id} - Q).$$

Each R_i has finite rank and the norms are uniformly bounded. If $w \in W$, then $Qw = w$ and $(\text{Id} - Q)w = 0$, so

$$R_iw = QA_iw \in W.$$

Finally, every $x \in E$ decomposes as $x = Qx + (\text{Id} - Q)x$, and

$$R_ix = QA_i(Qx) + (\text{Id} - Q)A_i((\text{Id} - Q)x) \longrightarrow Qx + (\text{Id} - Q)x = x$$

uniformly on compact sets, because $A_i \rightarrow \text{Id}_E$ uniformly on compact sets and $Q, \text{Id} - Q$ are bounded. $\square$

Counterexample

Theorem 1. *There is a separable Banach-space pair (X, Y) such that $(\mathcal{F}(X), \mathcal{F}(Y))$ has the BAP, but (X, Y) does not have the BAP.*

Proof. Use the pair constructed in Proposition 6.11 of the source paper. Namely, let Z be a separable Banach lattice without the approximation property, let (E_n) be a paving of Z , and set

$$X = c(E_n), \quad Y = c_0(E_n).$$

The source proves that (X, Y) has the Lipschitz BAP but not the BAP. It also records two further facts from the construction: X and Y have the BAP individually, and there is a Lipschitz retraction $P : X \rightarrow Y$.

Since X has the BAP, the Godefroy-Kalton theorem gives the BAP for $\mathcal{F}(X)$. Since P is a Lipschitz retraction fixing Y , its linearization

$$\widehat{P} : \mathcal{F}(X) \rightarrow \mathcal{F}(Y)$$

is a bounded projection: it maps $\mathcal{F}(X)$ into $\mathcal{F}(Y)$, and it restricts to the identity on $\mathcal{F}(Y)$. Applying the complemented-pair lemma to $E = \mathcal{F}(X)$, $W = \mathcal{F}(Y)$, and $Q = \widehat{P}$, we get that $(\mathcal{F}(X), \mathcal{F}(Y))$ has the BAP.

Thus $(\mathcal{F}(X), \mathcal{F}(Y))$ has BAP while (X, Y) does not. $\square$

Scope and Limitations

The result disproves the implication from BAP of the free-space Banach pair to BAP of the original Banach pair. It does not contradict the source paper’s positive characterizations of BAP for $(\mathcal{F}(M), \mathcal{F}(K))$ over compact or separable metric pairs, and it does not settle every possible formulation of a “similar result” for arbitrary metric pairs.

Novelty Check

The run indexes were searched for arXiv:1702.04774, the title, Ando-Choi-Effros lifting, BAP for pairs, Lipschitz-free spaces, and related phrases. No exact prior packet for this source question was found. External web searches on 2026-06-20 for phrases such as “BAP for pairs” with “Lipschitz-free spaces”, “ $F(X), F(Y)$ ” with “BAP” and “pair”, and “free-space pair” with “without the BAP” returned the source paper or no direct answer.

References

1. J. A. Chavez-Dominguez, arXiv:1702.04774, 2017.
2. G. Godefroy and N. J. Kalton, *Lipschitz-free Banach spaces*, Studia Math. 159 (2003), 121–141.

Human Review Recommendation

The main review point is interpretive: confirm that the source’s phrase “similar result” includes the natural Banach-pair implication $(\mathcal{F}(X), \mathcal{F}(Y)) \text{ BAP} \Rightarrow (X, Y) \text{ BAP}$. Under that reading, the counterexample is complete.